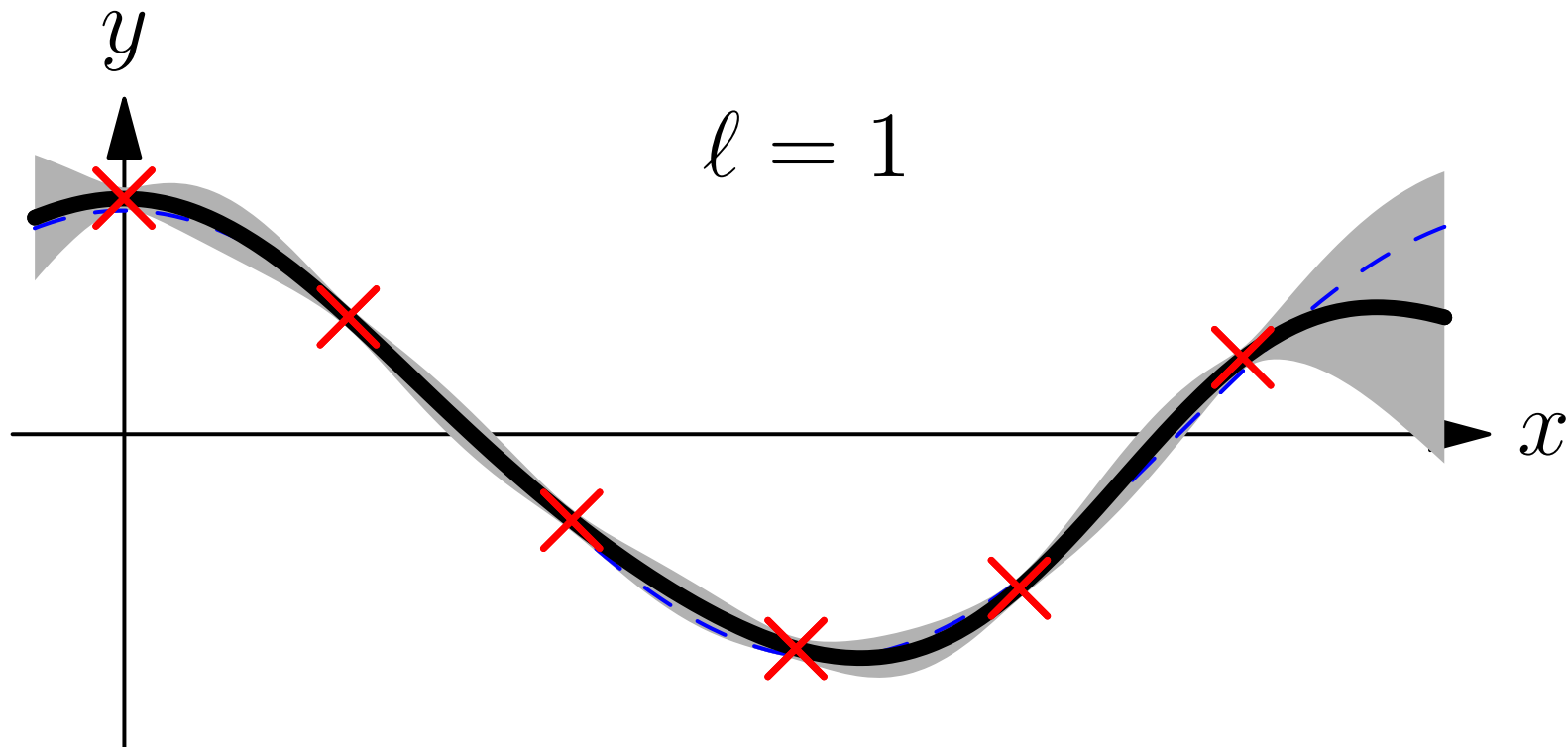


# Mathematics of Machine Learning

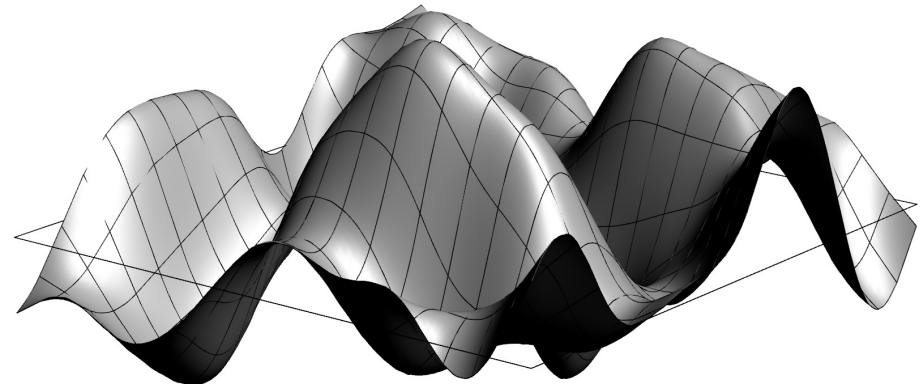
## *Gaussian Processes*



*Gaussian Processes, regression*

# Outline

1. **ML with Gaussians**
2. Gaussian Processes
3. Bayesian Inference
4. Hyper-parameters



# Machine Learning with Gaussians

- Due to the fact that we can analytically integrate over multivariate Gaussians they are heavily used of machine learning
- Examples include multivariate modelling, Kalman filters and Gaussian Processes
- Despite being analytically tractable, they are very fiddly
- This is mainly because it is easy to specify means,  $\mu$ , and covariance,  $\mathbf{C}$ , but Gaussians involve the inverse covariance matrix

$$\mathcal{N}(\mathbf{x} \mid \mu, \mathbf{C}) = \frac{e^{-\frac{1}{2}(\mathbf{x}-\mu)^{\top} \mathbf{C}^{-1}(\mathbf{x}-\mu)}}{\sqrt{\det(2\pi \mathbf{C})}}$$

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# Completing the Square

- One of the key tools for performing Gaussian integrals is “completing the square”

$$-\frac{1}{2}\mathbf{x}^\top \mathbf{C}^{-1} \mathbf{x} + \mathbf{a}^\top \mathbf{x} = -\frac{1}{2}\mathbf{x}^\top \mathbf{C}^{-1} \mathbf{x} + \frac{1}{2}\mathbf{a}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top \mathbf{a}$$

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- This gives us the Hubbard–Stratonovich transformation

$$\int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x} + \mathbf{a}^T\mathbf{x}}}{\sqrt{|\det(2\pi\mathbf{C})|}} d\mathbf{x} = \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\mathbf{x}^T - \mathbf{a}^T\mathbf{C})\mathbf{C}^{-1}(\mathbf{x} - \mathbf{C}\mathbf{a}) + \frac{1}{2}\mathbf{a}^T\mathbf{C}\mathbf{a}}}{\sqrt{|\det(2\pi\mathbf{C})|}} d\mathbf{x}$$

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# Moment Generating Function

- Again through completing the square we can show

$$M(\mathbf{a}) = \int_{-\infty}^{\infty} e^{\mathbf{a}^\top \mathbf{x}} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{C}) d\mathbf{x} = e^{\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{a}^\top \mathbf{C} \mathbf{a}}$$

- The first moment is given by

$$\mathbb{E}[\mathbf{x}] = \nabla_{\mathbf{a}} M(\mathbf{a}) \big|_{\mathbf{a}=\mathbf{0}} = (\boldsymbol{\mu} + \mathbf{C} \mathbf{a}) e^{\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{a}^\top \mathbf{C} \mathbf{a}} \big|_{\mathbf{a}=\mathbf{0}} = \boldsymbol{\mu}$$

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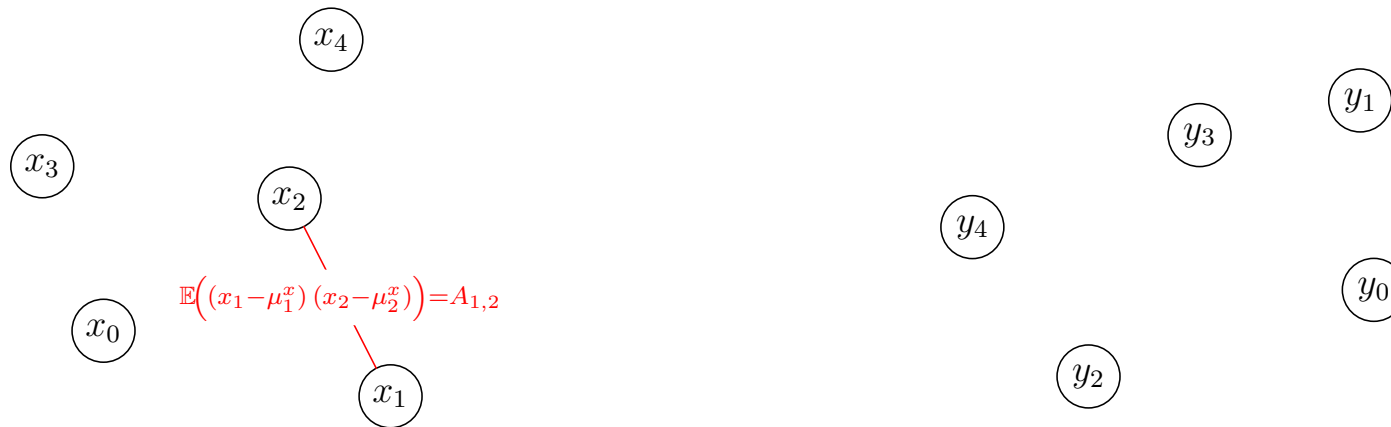
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# Multivariate Normals



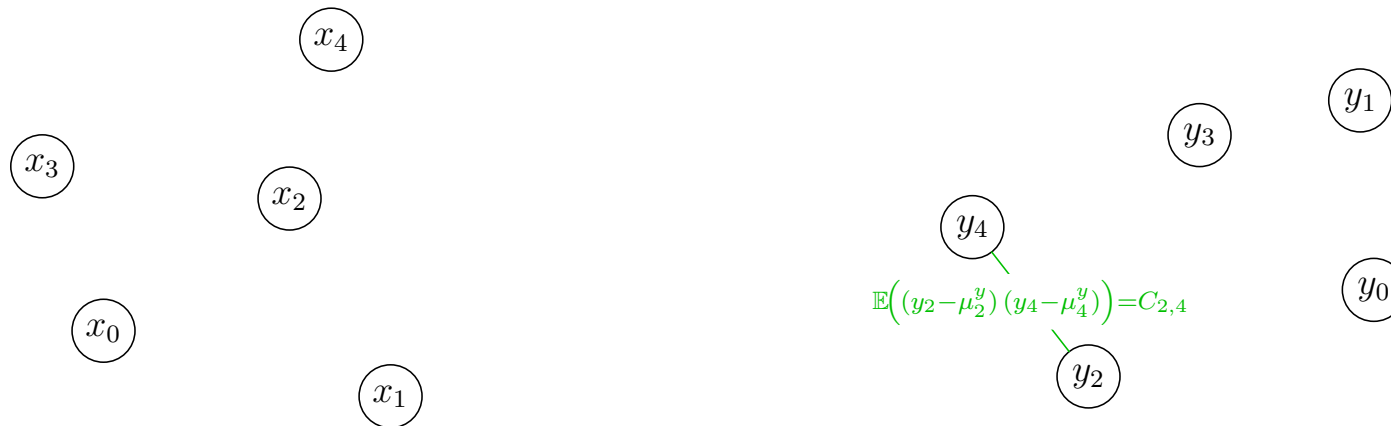
$$\mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \boldsymbol{\mu}^x \\ \boldsymbol{\mu}^y \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}\right) = \frac{e^{-\frac{1}{2}(\mathbf{x}^\top - \boldsymbol{\mu}^{x\top}, \mathbf{y}^\top - \boldsymbol{\mu}^{y\top}) \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x} - \boldsymbol{\mu}^x \\ \mathbf{y} - \boldsymbol{\mu}^y \end{pmatrix}}}{\sqrt{\det(2\pi \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix})}}$$

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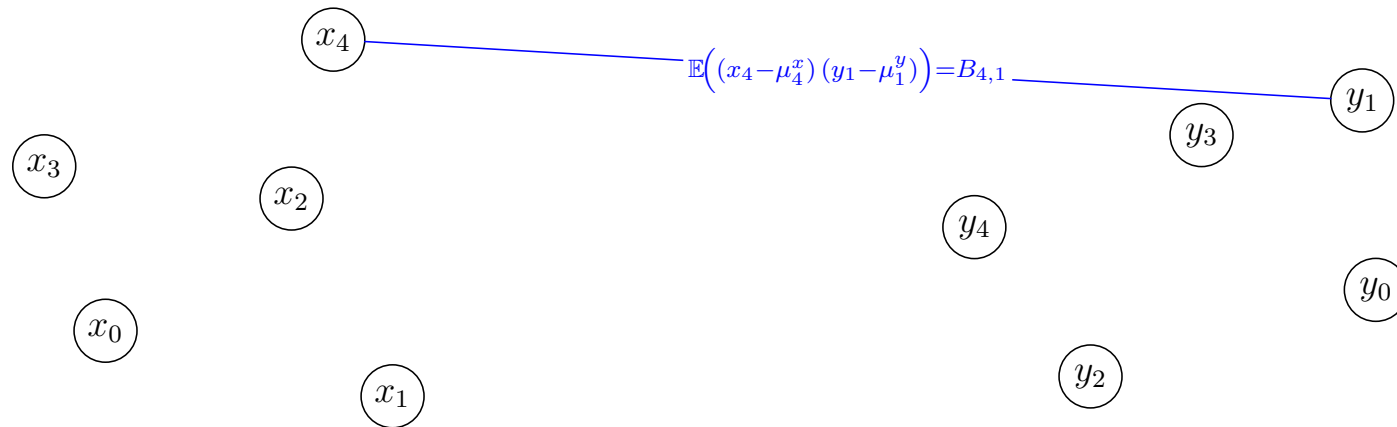
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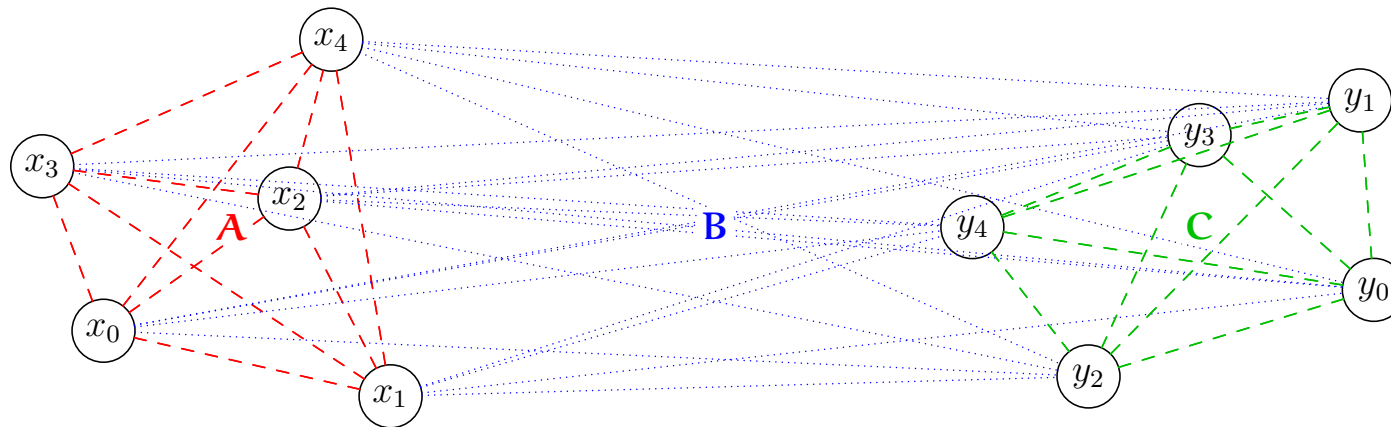
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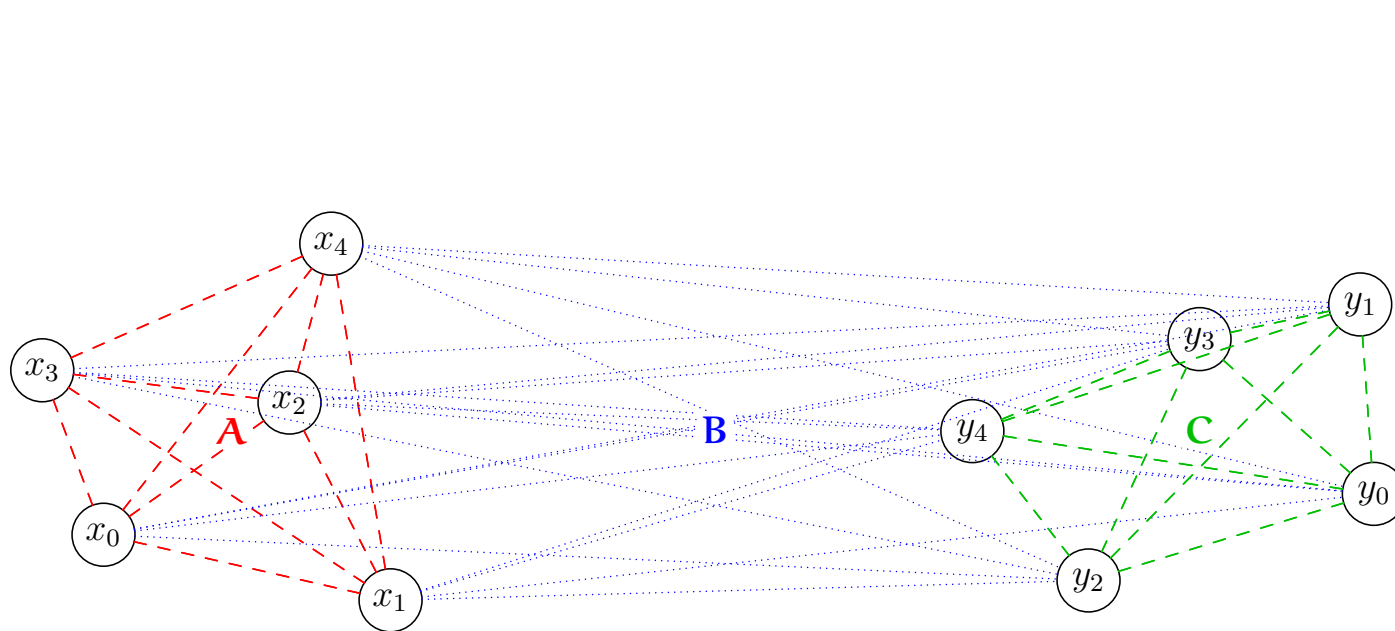
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# Marginalising Multivariate Normals

- I can define a moment generating function,

$$\begin{aligned} M(\mathbf{a}, \mathbf{b}) &= \int e^{(\mathbf{a}^\top, \mathbf{b}^\top) \begin{pmatrix} \mathbf{x} - \boldsymbol{\mu}^x \\ \mathbf{y} - \boldsymbol{\mu}^y \end{pmatrix}} \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \boldsymbol{\mu}^x \\ \boldsymbol{\mu}^y \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}\right) d\mathbf{x} d\mathbf{y} \\ &= e^{\frac{1}{2}(\mathbf{a}^\top, \mathbf{b}^\top) \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}} = e^{\frac{1}{2}(\mathbf{a}^\top \mathbf{A} \mathbf{a} + \mathbf{b}^\top \mathbf{C} \mathbf{b} + 2\mathbf{a}^\top \mathbf{B} \mathbf{b})} \end{aligned}$$

- If I set  $\mathbf{b} = \mathbf{0}$  this is equivalent to marginalising out  $\mathbf{y}$
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- An important and non-trivial result is

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- You would think this is easy to prove by straightforward integration
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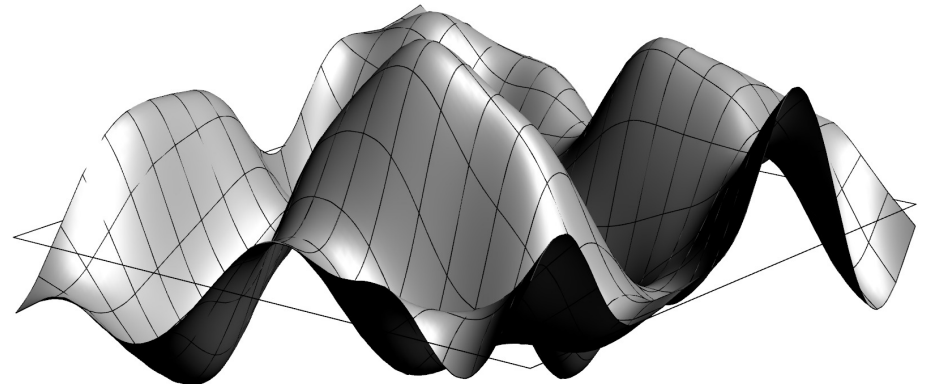
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# Outline

1. ML with Gaussians
2. **Gaussian Processes**
3. Bayesian Inference
4. Hyper-parameters



# Gaussian Processes

- Gaussian processes (GPs) are a mathematically defined ensemble of functions
- They can be combined with Bayesian inference to give one of the most powerful regression techniques
- Although Bayesian they can be used in a black-box fashion due to the ubiquity of the prior
- Mathematically they are a bit complicated
- In practice they aren't that difficult to use

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# Regression

- In regression we try to fit a multi-dimensional function to our data
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# Priors on Functions

- We can think of a solution as a function  $f(\mathbf{x})$
- We can put a prior probability distribution,  $p(f)$ , on a function,  $f$ , that prefers smooth functions
- We can then compute a posterior probability distribution on functions given the data,  $p(f|\mathcal{D})$
- As a likelihood,  $p(y_i|f(\mathbf{x}_i))$ , we use the probability of observing  $y_i$  given the true function value is  $f(\mathbf{x}_i)$
- In general, this would be next to impossible to compute

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# Gaussian Processes

- Gaussian Processes are probability distributions over functions
- (Functions can be viewed as vectors in an infinite dimensional vector space)
- In the Gaussian Process,  $\mathcal{GP}(m, k)$ , the probability of a function,  $f$ , is proportional

$$p(f|m, k) \propto e^{-\frac{1}{2} \int (f(\mathbf{x}) - m(\mathbf{x})) k^{-1}(\mathbf{x}, \mathbf{y}) (f(\mathbf{y}) - m(\mathbf{y})) d\mathbf{x} d\mathbf{y}}$$

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# Meaning of GP

- To understand GP's we can discretise space,  $\mathbf{x}$ , into a lattice of points  $\{\mathbf{x}_i\}$
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# Covariance function

- $k(\mathbf{x}, \mathbf{y})$  is a covariance function

$$\mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{y}) - m(\mathbf{y}))] = k(\mathbf{x}, \mathbf{y})$$

- This is sometimes know as a **kernel**—it must be positive semi-definite
- It is a free “parameter” that the user gets to choose (although we can learn its parameters too)
- If  $k(\mathbf{x}, \mathbf{y})$  is a function of  $\mathbf{x} - \mathbf{y}$  it is “**stationary**”
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# Covariance function

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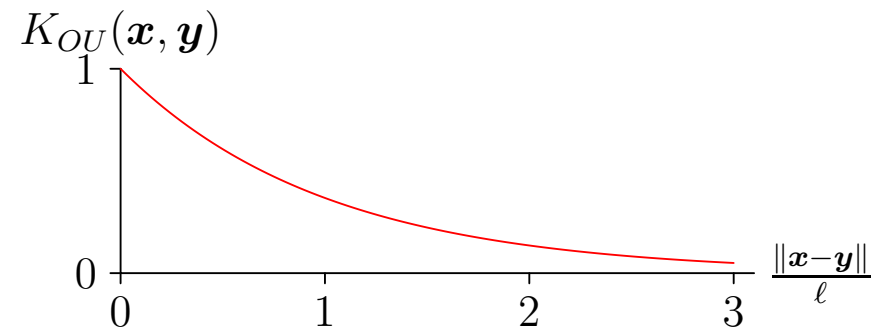
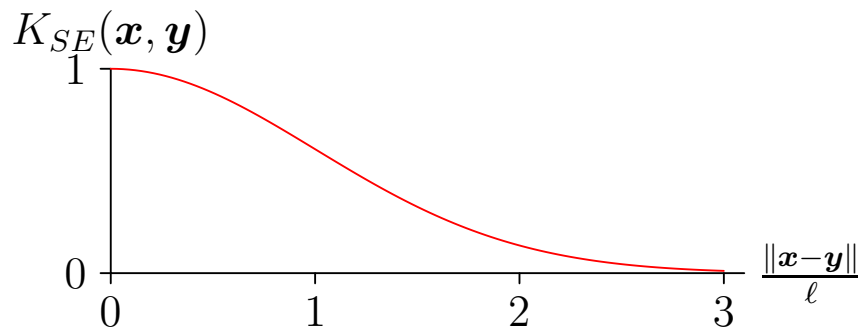
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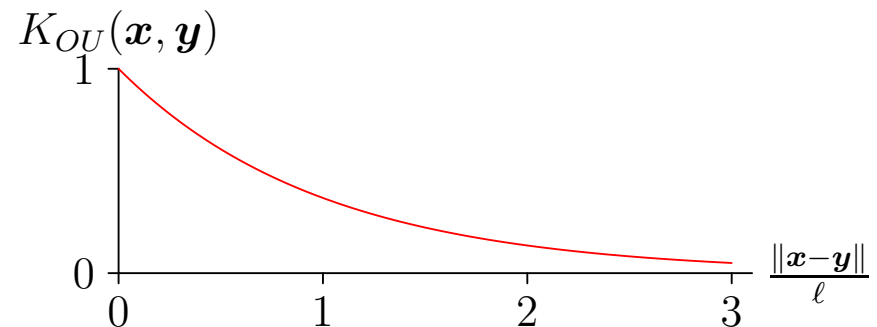
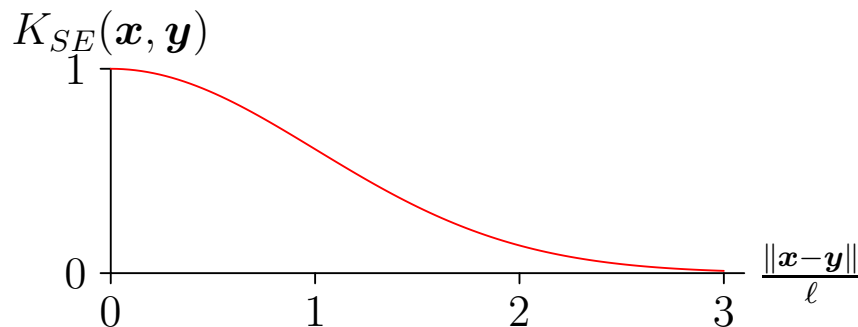
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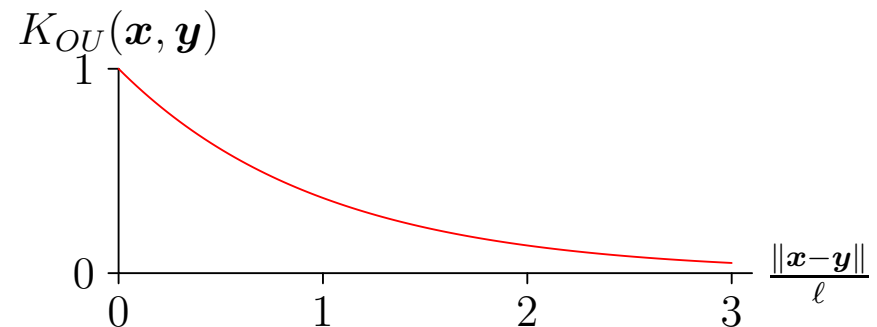
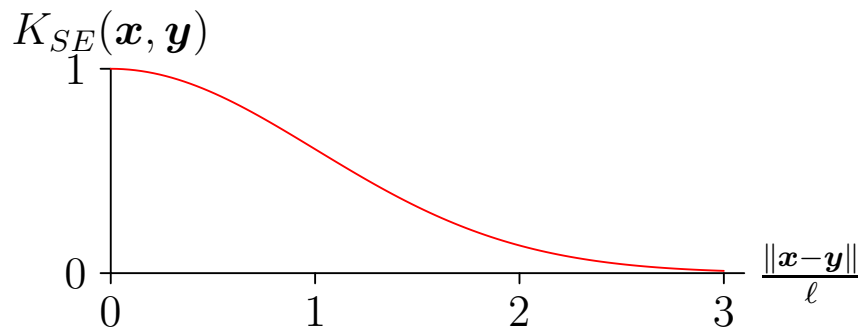
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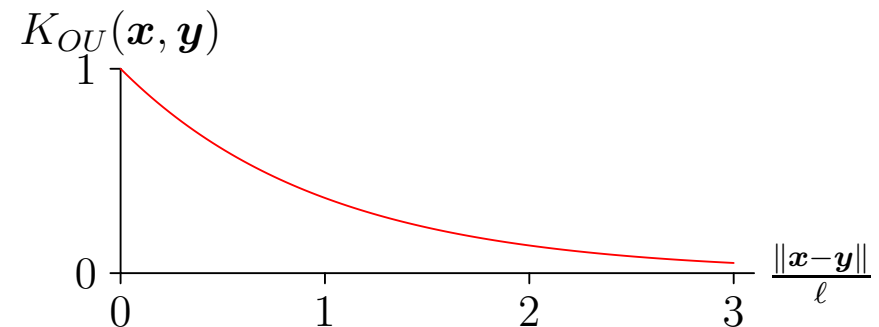
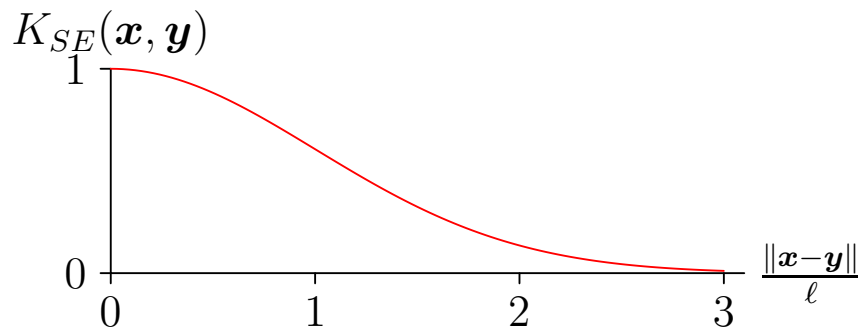
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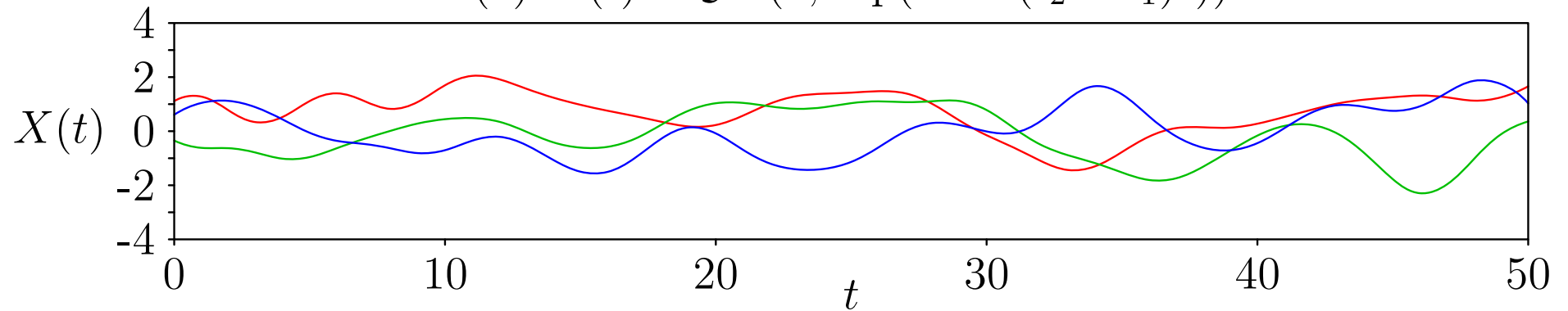
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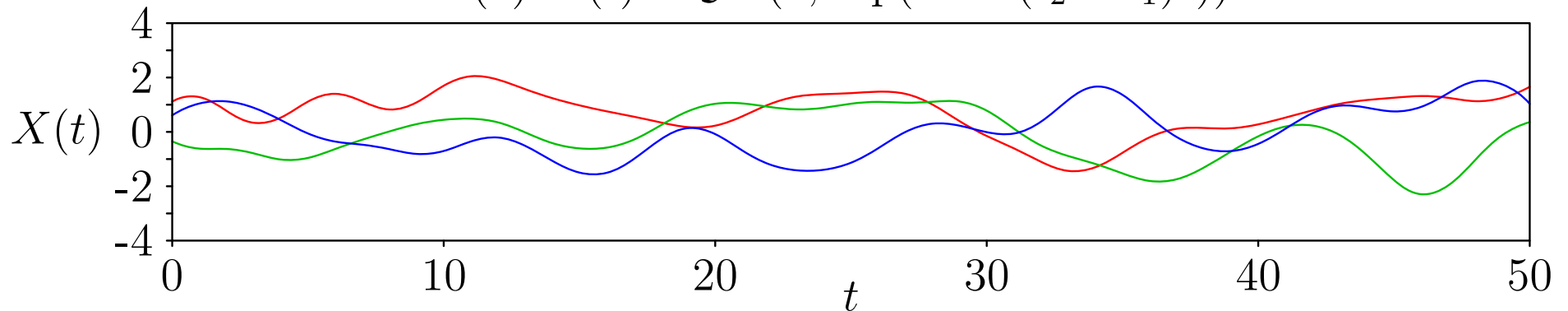
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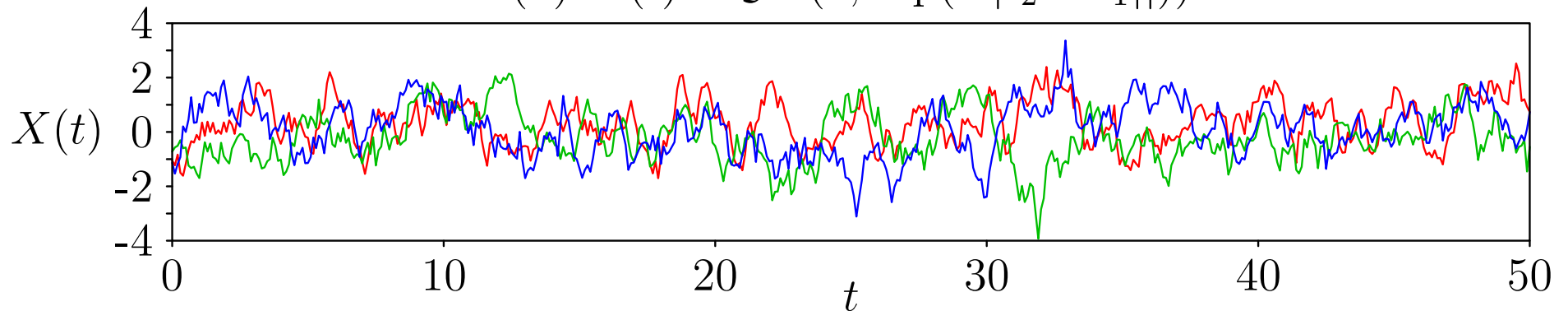


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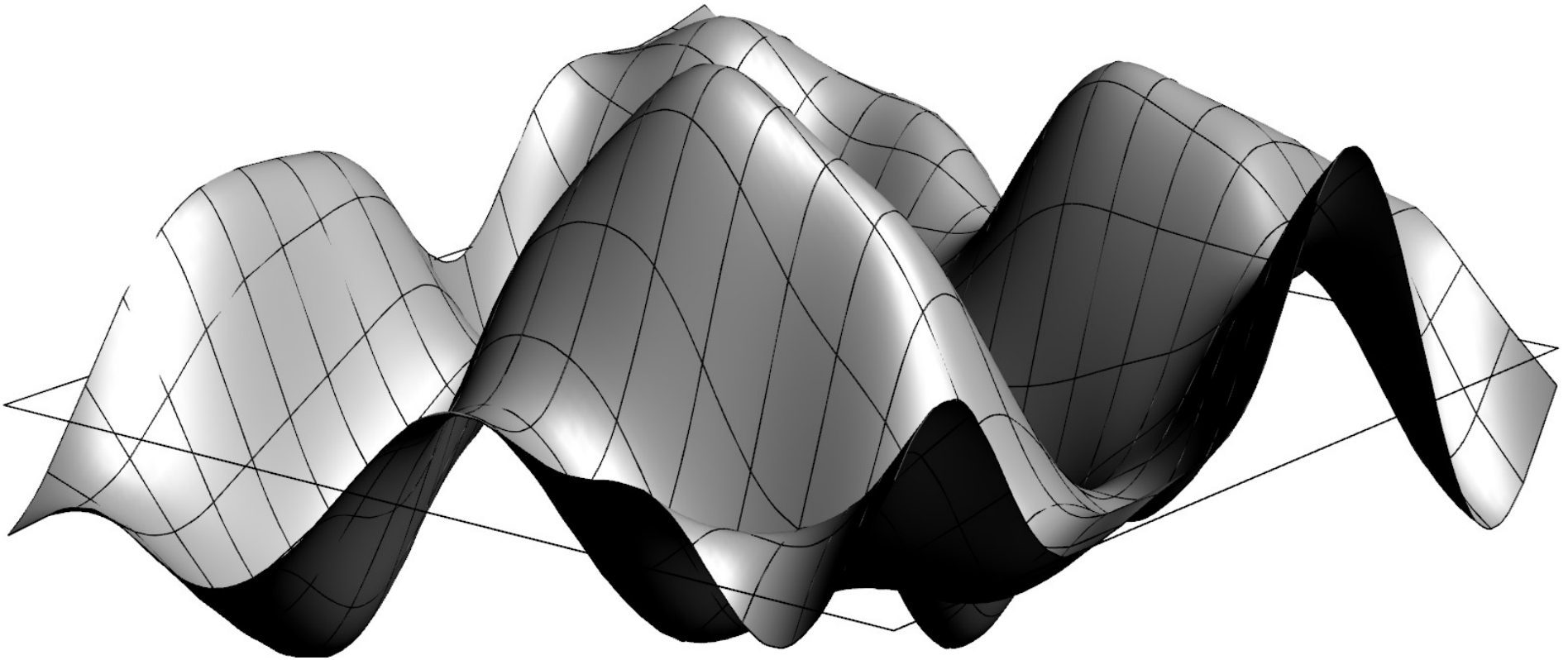
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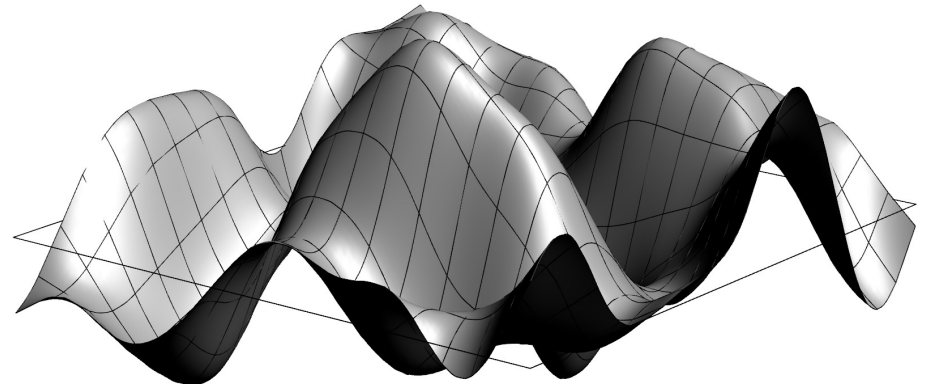


# 2-D Gaussian Processes



# Outline

1. ML with Gaussians
2. Gaussian Processes
3. **Bayesian Inference**
4. Hyper-parameters



# Observed Gaussian Processes

- Given some data points  $\mathcal{D} = ((\mathbf{x}_i, y_i) | i = 1, \dots, m)$  the likelihood (assuming Gaussian error are independence of the data point) is given by

$$p(\mathcal{D}|f) = \prod_{i=1}^m \mathcal{N}(y_i | f(\mathbf{x}_i), \sigma^2)$$

- Using a Gaussian Process prior we can compute a posterior using Bayes's rule
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# Alternative Derivation

- Denoting the target values as a vector  $\mathbf{y}$  with elements  $y_i$
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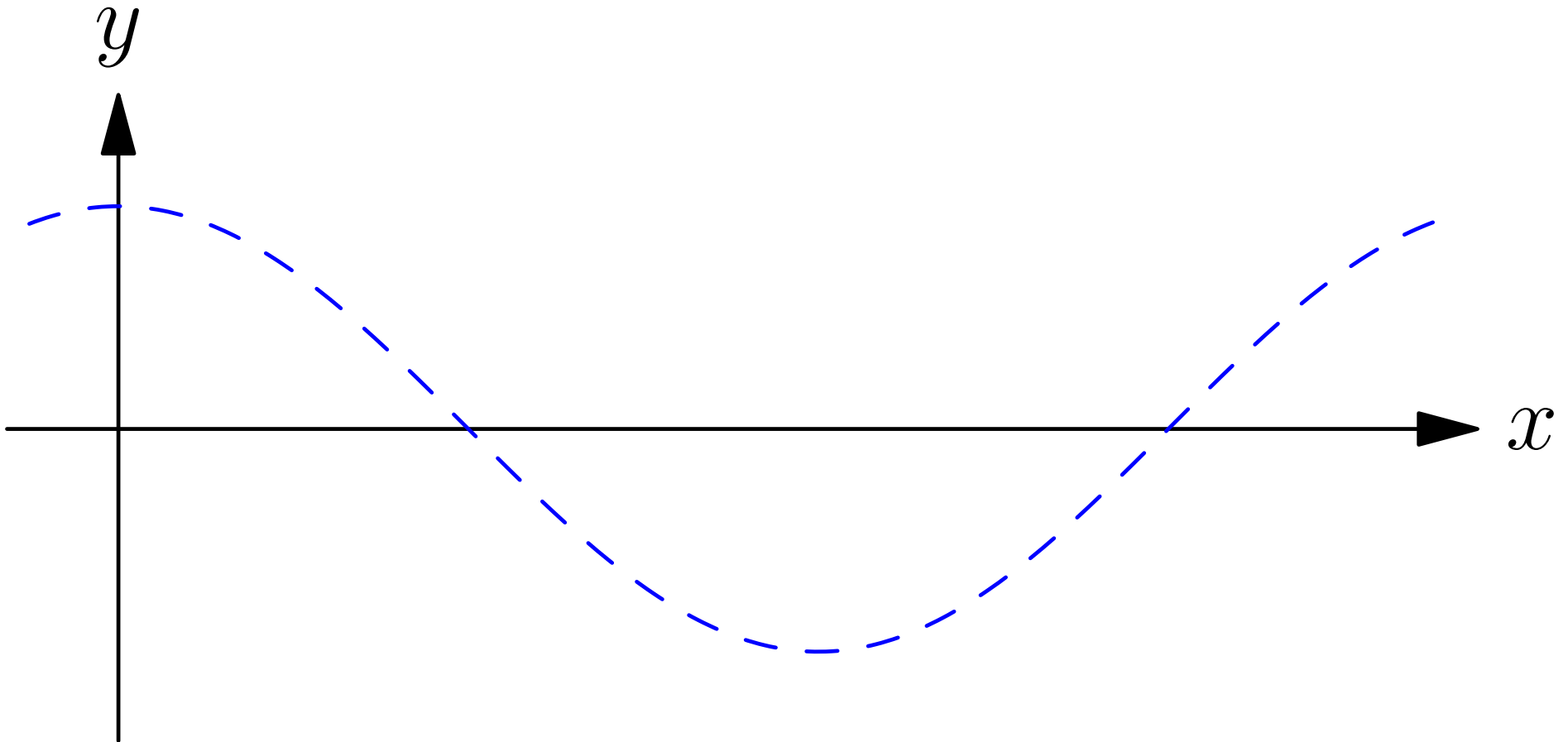
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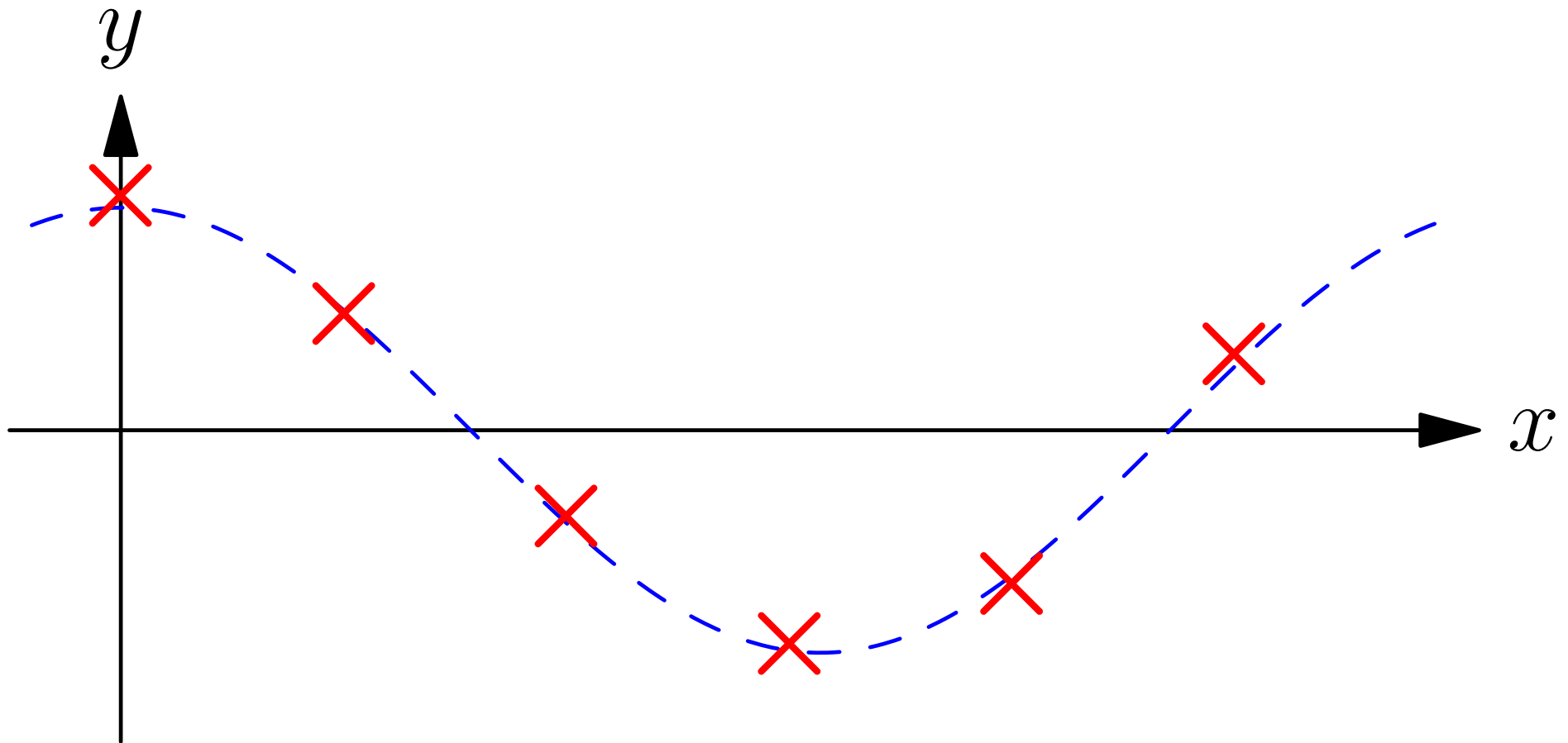
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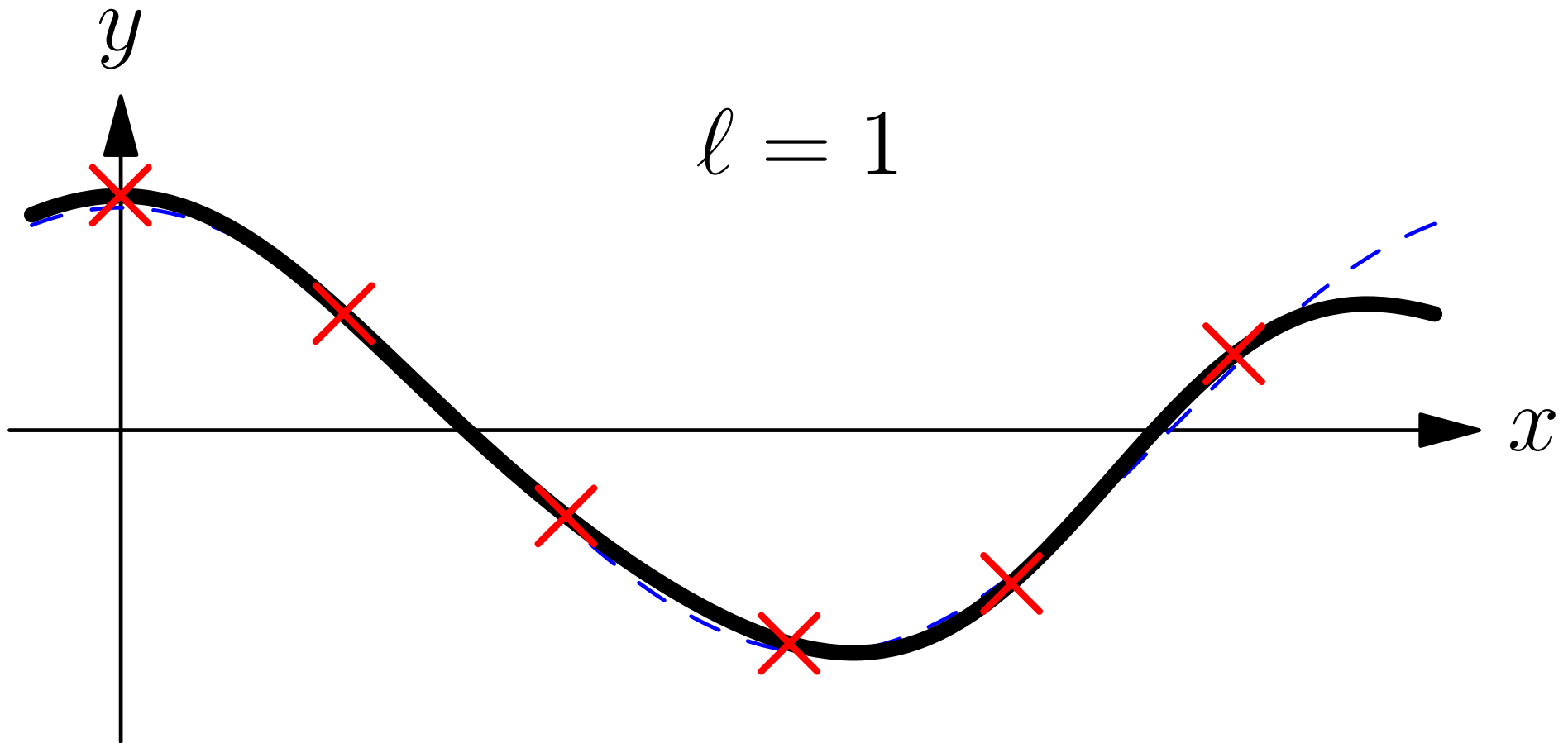


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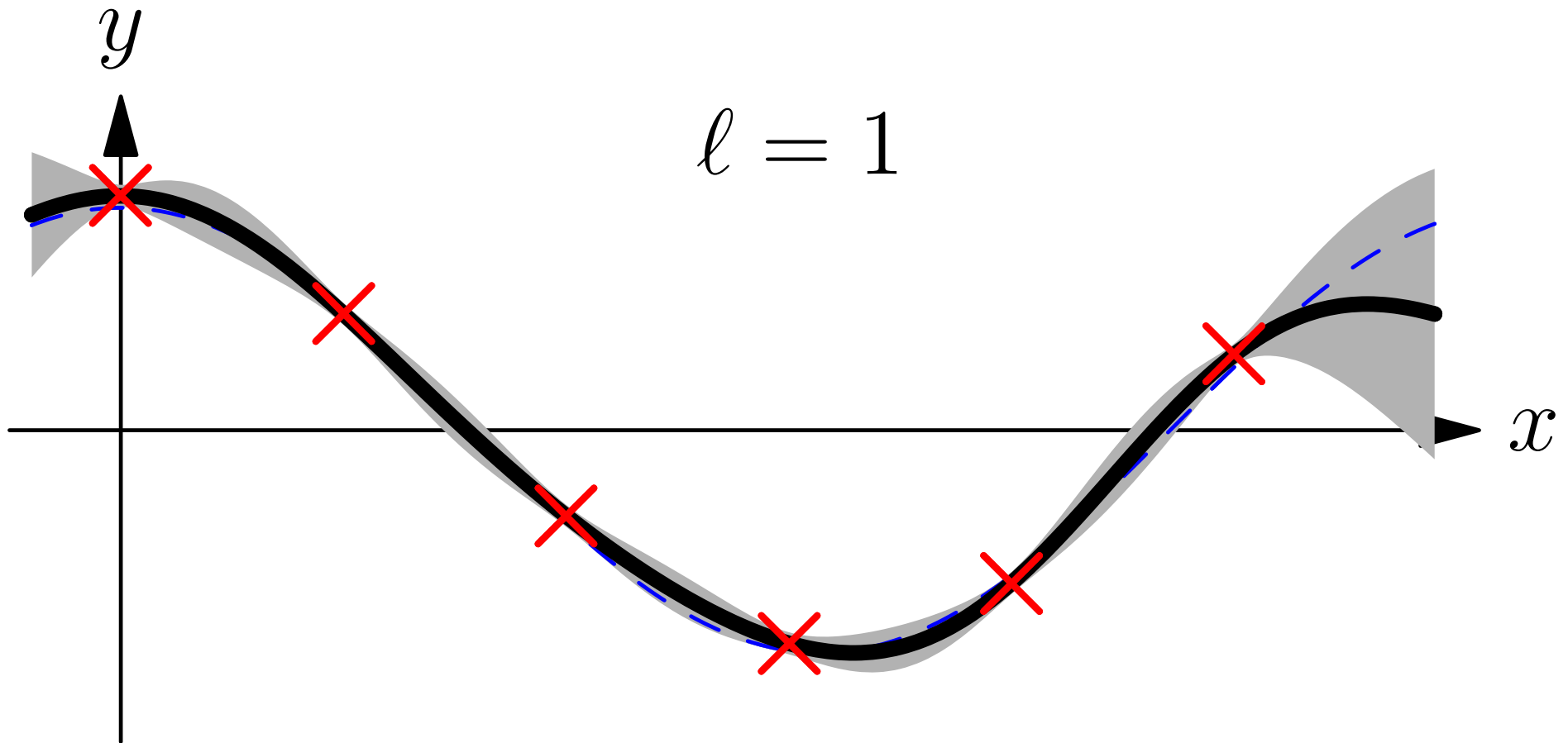
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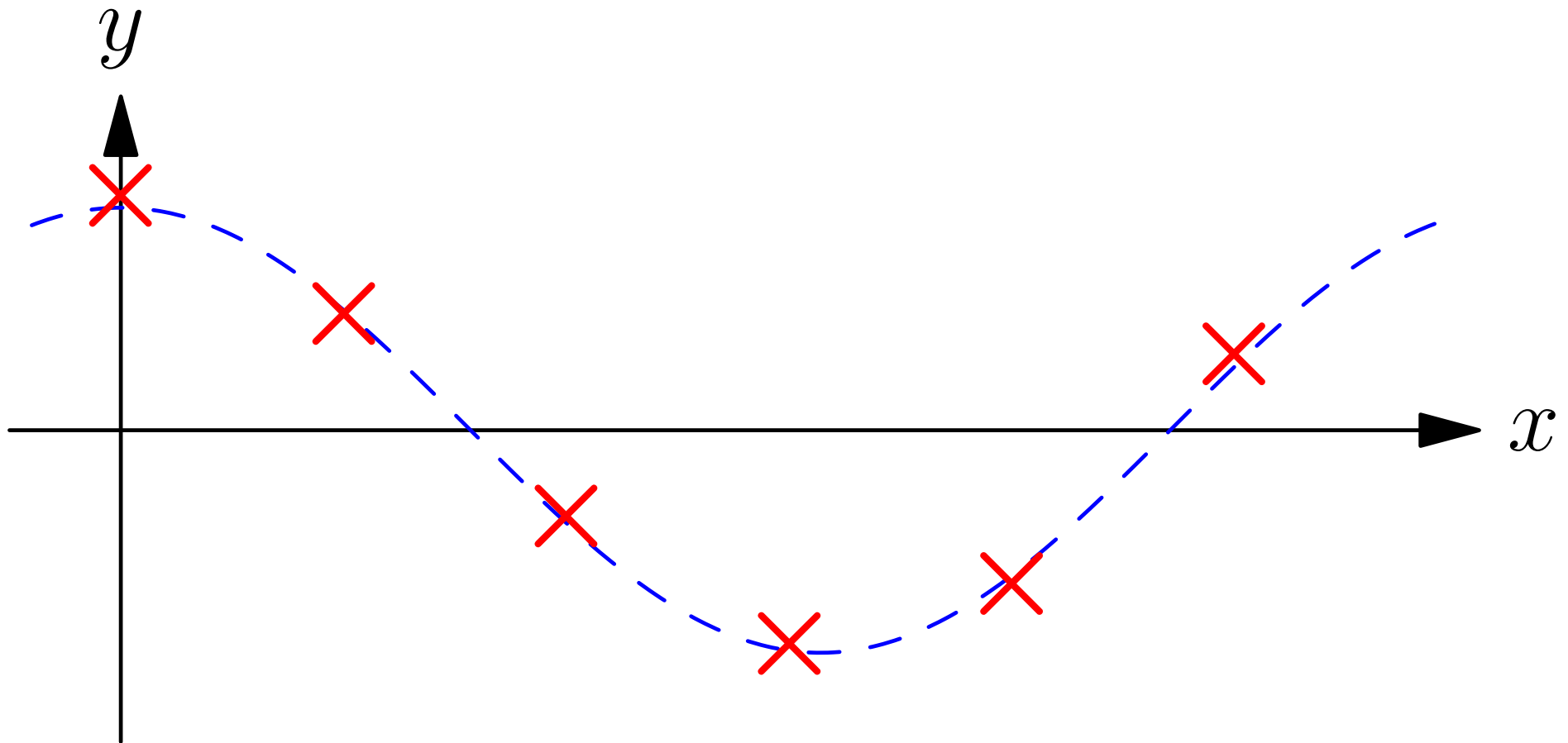


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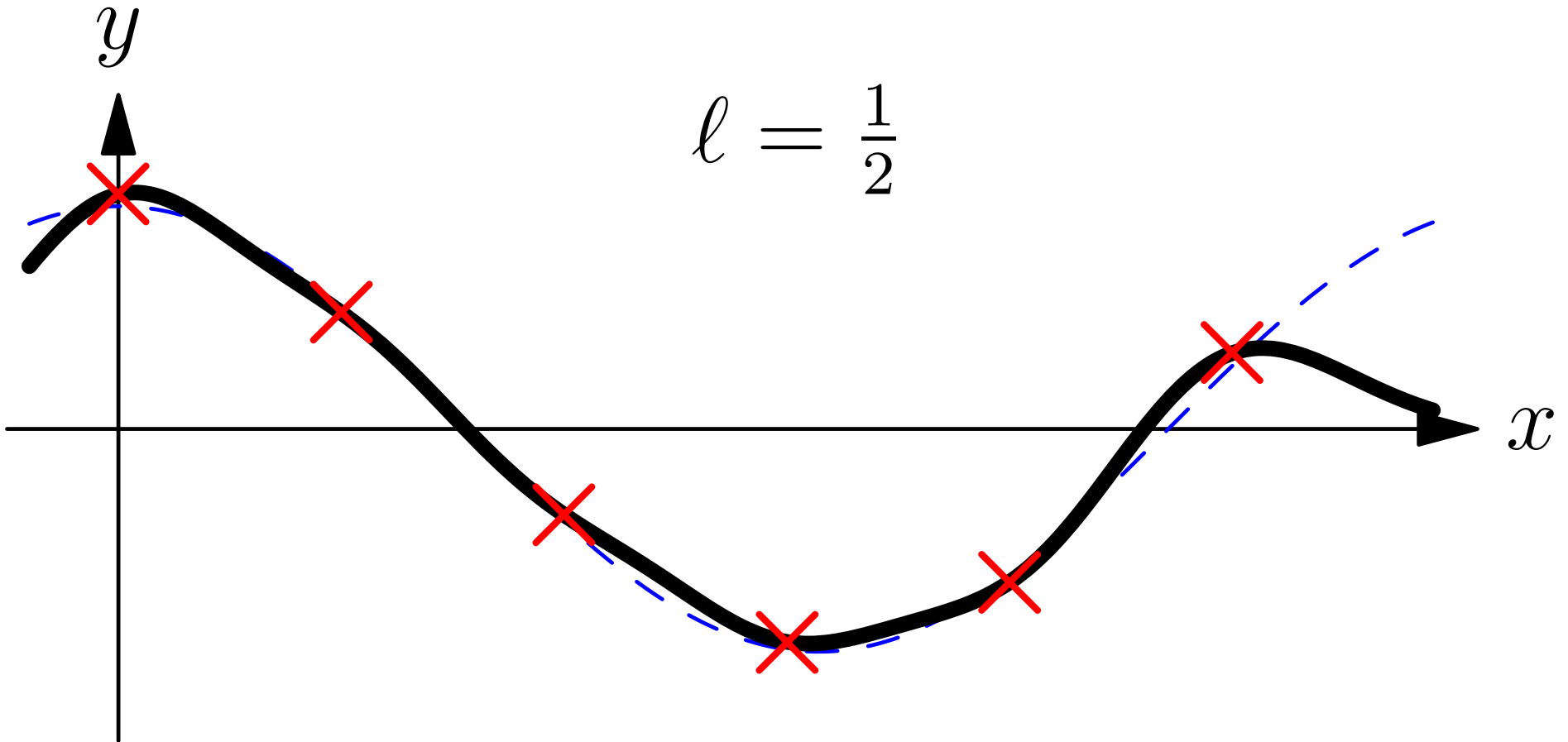


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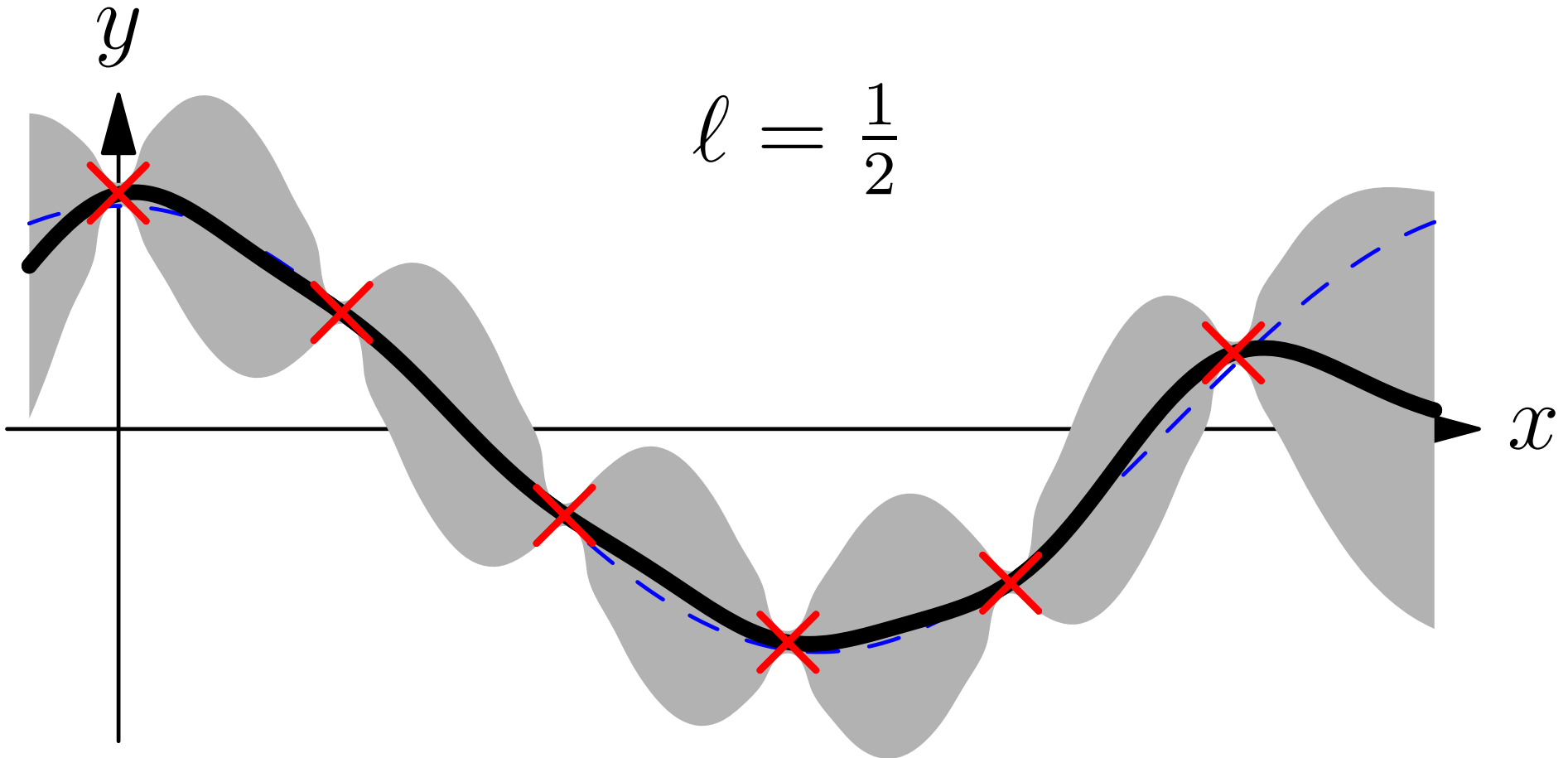
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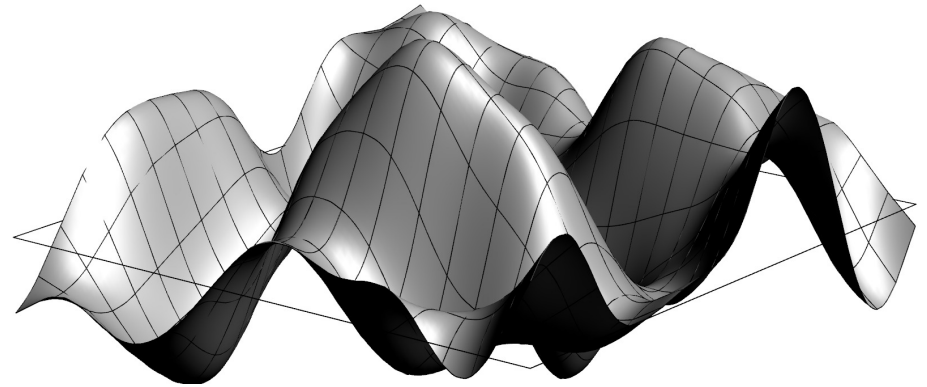
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# Outline

1. ML with Gaussians
2. Gaussian Processes
3. Bayesian Inference
4. **Hyper-parameters**



# Choosing the Correct Covariance Function

- Choosing the correct covariance function is critical
- Most covariance functions include a continuous **hyper-parameter** (e.g. the correlation length  $\ell$ ) that we have to choose correctly
- This is typical of many Bayesian problems where we have some set of hyper-parameters,  $\phi$ , describing the model
- These are different to the normal parameters we learn (e.g. weights  $w$  or in GP the functions  $f(x)$ )
- In Bayesian inference we learn the posterior for these normal parameters

$$p(f|\mathcal{D}, \phi) = \frac{p(\mathcal{D}|f, \phi)p(f|\phi)}{p(\mathcal{D}|\phi)}$$

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# Evidence Framework

- The normalisation factor,  $p(\mathcal{D}|\phi)$  is known as the **marginal likelihood** or **evidence**

$$p(\mathcal{D}|\phi) = \int p(\mathcal{D}|f, \phi) p(f|\phi) \mathrm{d}f$$

- We can perform a Bayesian calculation at a second level by putting a prior on  $\phi$

$$p(\phi|\mathcal{D}) = \frac{p(\mathcal{D}|\phi)p(\phi)}{p(\mathcal{D})}$$

- From this we can now marginalise out the hyper-parameters

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# Evidence Framework

- The normalisation factor,  $p(\mathcal{D}|\phi)$  is known as the **marginal likelihood** or **evidence**

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# Maximum-Likelihood-II

- The integral

$$p(f|\mathcal{D}) = \int p(f|\mathcal{D}, \phi) p(\phi|\mathcal{D}) d\phi$$

usually can't be computed analytically and we have to use Monte Carlo methods (see later lecture)

- An alternative is to use the most likely hyper-parameter
- We can find this by using gradient search of  $p(\mathcal{D}|\phi)$
- This is sometimes referred to as ML-II
- Normally even this can be difficult, but for GP its not too difficult

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# Evidence for GP

- For GP the (log)-evidence can be computed in closed form

$$\log(p(\mathcal{D}|\phi)) = -\frac{1}{2}\mathbf{y}^\top(\mathbf{K} + \sigma^2\mathbf{I})\mathbf{y} - \frac{1}{2}\log(|\mathbf{K} + \sigma^2\mathbf{I}|) - \frac{m}{2}\log(2\pi)$$

- ★ First term measures goodness of fit
  - ★ Second term measure complexity of model
  - ★ Last term is a common normalisation constant
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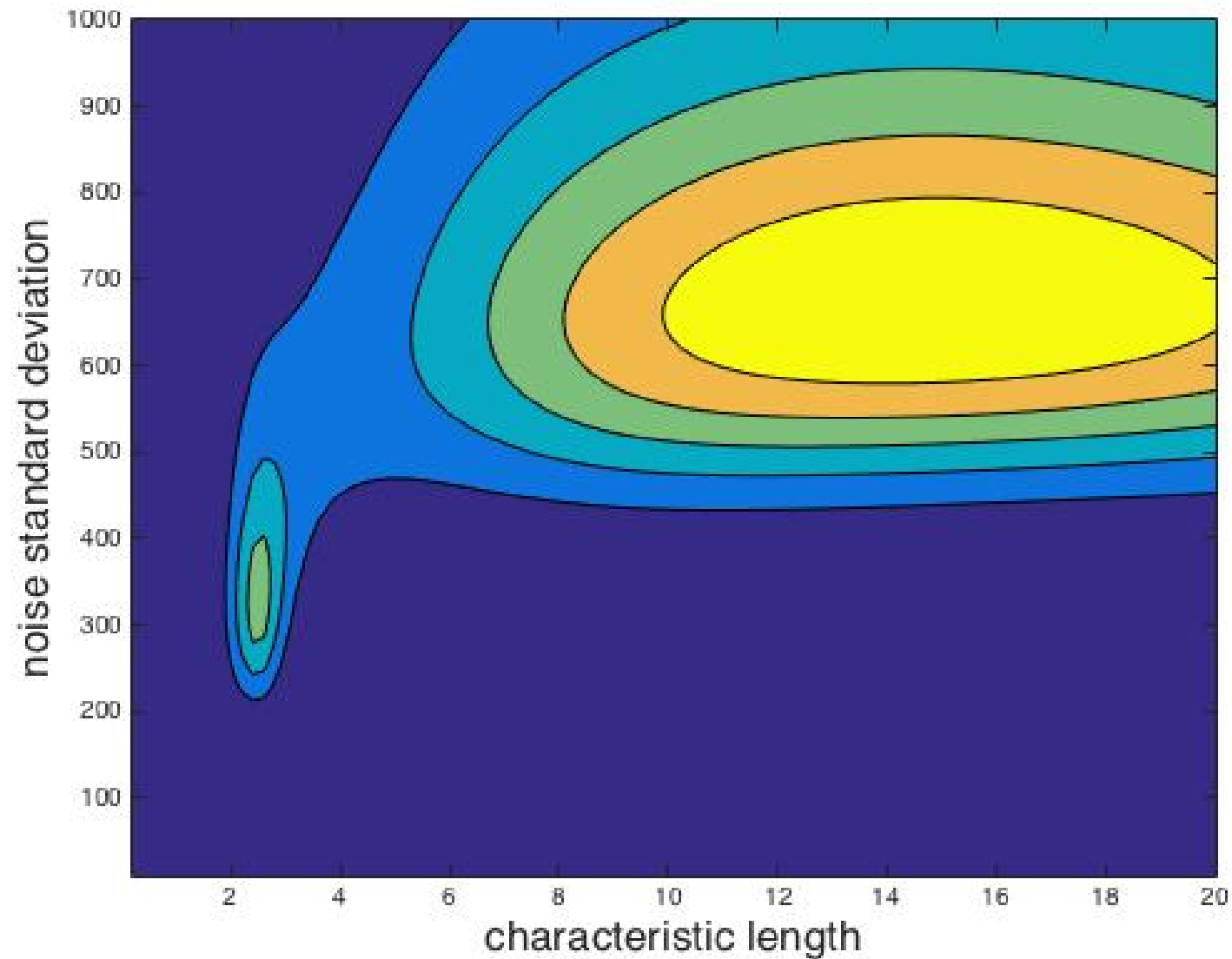
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- Can efficiently compute derivatives and find best parameters
  - Could overfit!

# Example (slightly pathological)



# Conclusions

- Gaussian processes are very powerful for regression (and classification?)
- Because all calculations involve Gaussian integrals we can compute everything in closed form
- (Actually its a pain to do the mathematics because you end up working with inverse of matrices)
- Fairly generic (black-box) technique because the prior captures many continuity constraints
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